

JAGANATH MK

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PROFESSIONAL SUMMARY

B.Tech graduate with a strong foundation in Python programming, SQL, Django and core computer science concepts, including data structures and problem-solving. Hands-on experience in developing software-based projects that demonstrate logical thinking, debugging skills, and clean coding practices. Proficient in writing efficient, maintainable code and working with databases to build reliable applications. A quick learner with a passion for software development and continuous improvement, seeking full-time opportunities to contribute to scalable, real-world solutions in a professional software environment.

EDUCATION

B.Tech in Electronics and Communication Engineering

College of Engineering, Thalassery

2021 – 2025

Thalassery, Kerala

PROJECTS

Anti -Theft Flooring System [↗](#)

- Developed an IoT-based smart flooring system for real-time intrusion detection using ESP32 and piezoelectric sensors.
- Integrated GSM SIM800L module to send SMS alerts for immediate notification of unauthorized access.
- Used a buzzer to emit audible alarms, ensuring quick awareness of security breaches.
- Utilized ESP32 microcontroller to efficiently process sensor data and control system components.
- Designed the system for power efficiency with support for both battery and external power sources.
- Optimized algorithms to reduce false alarms and improve detection accuracy.
- Technologies used: ESP 32 Microcontroller, piezoelectric sensors, GSM SIM800L module, buzzer.
- Provided a cost-effective and scalable security solution with potential applications in homes, warehouses, and commercial establishments.

Smart-Trolley With Auto Bill Payment System [↗](#)

- Developed an intelligent smart trolley system using ESP32 and YOLOv8 for real-time item detection and automated billing.
- Built a Django-based backend to manage detected items, pricing, and billing logic efficiently.
- Designed a web interface to display item details, prices, and total cost directly on a centralized screen.
- Enabled real-time synchronization between the trolley system and the billing interface.
- Designed the system to autonomously track and update the shopping cart, minimizing human interaction.
- Technologies used: ESP32, QR scanner, YOLOv8, Django(backend), Python, OpenCv.
- Enhanced customer experience by eliminating queues and reducing manual intervention and providing a smooth, automated shopping flow.

SKILLS

Languages: Python Programming

Web Framework: Django

Database: MySQL

Soft Skills: Communication Skills, Teamwork, Problem-Solving

CERTIFICATIONS

- Problem Solving (Basic) Certificate - Hackerrank
- MySQL Explorer Badge - Oracle